

Heart Failure: Hospital-at-Home

HaH: M-190-HaH (ISC)

[Link to Codes](#)

This MCG guideline has not yet been updated to reflect new guidance and requirements found in the 2024 Medicare Advantage and Part D Final Rule (CMS-4201-F). For Medicare Advantage beneficiaries, before referring to MCG's guidelines users should first follow all guidance related to this topic located in CMS National Coverage Determinations (NCDs), Local Coverage Determinations (LCDs), and other statutes and regulations. Content used in MCG guidelines may be used to supplement but does not replace, modify, or supersede existing Medicare regulations or applicable NCDs or LCDs.

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Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Admission to Hospital-at-Home

Note: Some patients may be appropriate for observation care. For consideration of observation care, see Heart Failure: Observation Care [ISC](#).

Note: For patients requiring hospital-based inpatient care rather than hospital-at-home, see Heart Failure [ISC](#).

- Admission for adult to Hospital-at-Home is indicated by **ALL** of the following (1)(2)(3)(4)(5)(6)(7)(8)(9)(10):[N](#)
 - Inpatient level of care needed, as indicated by **1 or more** of the following:
 - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
 - ☐ Pulmonary edema that is persistent, as indicated by **ALL** of the following:
 - New need for oxygen therapy to keep oxygen saturation above 90% (or increased FiO₂ need from baseline)
 - Has not improved sufficiently with observation care (eg, appropriately dosed IV diuretics)^[A]
 - Tachypnea or dyspnea (above baseline) that persists despite observation care
 - ☐ Severity of illness^[B] (upon presentation or after observation care) appropriate for Hospital-at-Home, as indicated by **ALL** of the following:

- Hypotension absent
- ☐ Oxygenation sufficient, as indicated by **1 or more** of the following:
 - Supplemental oxygen not needed
 - FiO₂ less than 45% (6 L/min) required to keep oxygen saturation (SpO₂) at 90% or greater
- No evidence of inadequate perfusion (eg, metabolic acidosis, Altered mental status, myocardial ischemia)
- Renal function stable (no ongoing decline)
- Absence of severe peripheral edema (eg, impairing ability to void, ambulate)
- ☐ Patient care needs (upon presentation or after observation care) appropriate for Hospital-at-Home, as indicated by **ALL** of the following:
 - No new need for invasive or noninvasive ventilatory assistance
 - No need for invasive monitoring (eg, pulmonary artery catheter, arterial line) or immediate response telemetry monitoring (ie, dangerous arrhythmia detection and subsequent immediate intervention)[C]
 - Medication regimen not beyond capabilities of Hospital-at-Home (eg, dosing interval)
- ☐ Patient medical history and comorbidities appropriate for Hospital-at-Home, as indicated by **ALL** of the following:
 - No known severe cardiac valvular disease (eg, aortic stenosis, mitral regurgitation)
 - Patient does not meet inpatient admission criteria for a second diagnosis or have care needs beyond capabilities of Hospital-at-Home due to an active clinically significant comorbidity

Alternatives to Admission

- Alternatives include(1)(2)(3)(4)(23)(24)(25)(26)(27):[\[B\]](#)
 - For arrhythmia: emergency department or rapid treatment site intervention(28):
 - Pharmacologic therapy to slow supraventricular arrhythmias
 - Cardioversion (electrical or pharmacologic) to sinus rhythm
 - For heart failure with pulmonary congestion or peripheral edema:
 - Outpatient therapy (eg, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), diuretics, beta-blocker, aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide), patient education, salt and fluid restriction)[D](1)(27)
 - Multidisciplinary team care with home care support(1)(30)
 - Emergency department or rapid treatment site care with oxygen, respiratory therapy, IV diuretics, correction of fluid and electrolyte balance, and control of arrhythmias(23)(31)(32)
 - Recovery facility care for ongoing parenteral medication, oxygen therapy, weight and fluid balance, laboratory testing, electrolyte correction, and management of comorbidities
 - Skilled nursing facility for medication adjustment, weight and fluid balance, nursing assessment, and nutritional support(33)
 - Observation
 - Admission to inpatient facility. See Heart Failure [\[B\]](#) ISC guideline.

Alternative Care Planning

- Care planning needs for patient not requiring admission may include(1)(3)(27)(34)(35)(36)(37):
 - Diagnostic tests, including(38):
 - Laboratory testing, including CBC, electrolytes, fasting glucose, glycosylated hemoglobin, kidney function, urinalysis, liver function, lipid profile, transferrin, HIV, and thyroid function
 - Cardiac biomarkers (eg, BNP, N-terminal pro-BNP level, and Troponin)(39)(40)
 - Vitamin D level, iron studies(41)
 - Chest x-ray
 - Echocardiogram
 - Chest ultrasound(42)
 - ECG
 - Oxygen saturation
 - Radionuclide ventriculography
 - Cardiac MRI
 - Ventricular viability testing (eg, nuclear study or cardiac MRI, for patients with ischemic cardiomyopathy)(43)
 - Pulmonary function tests
 - Pulmonary CT angiogram or lung ventilation perfusion scan
 - Treatments and procedures, including:
 - Pharmacologic treatment (eg, diuresis, beta-blockers, renin-angiotensin inhibitors, aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors)[A][D][E](1)(27)(35)(45)
 - Antihypertensive medication[F](29)
 - Cardioversion (electrical or pharmacologic) for supraventricular tachycardia or atrial fibrillation(28)
 - Consultation, assessment, and other services, including:
 - Cardiology consultation

- Electrophysiology specialist consultation (eg, arrhythmia aspect of acute presentation, resynchronization therapy, cardioverter-defibrillator placement).(1)(46)(47)(48)(49) See Electrophysiologic Study and Intracardiac Catheter Ablation [ISC](#), Left Atrial Appendage Closure, Percutaneous [ISC](#), or Electrophysiologic Study and Implantable Cardioverter-Defibrillator (ICD) Insertion [ISC](#) as appropriate.
- Smoking cessation counseling(2)
- Nutrition counseling(1)(29)
- Cardiac rehabilitation(50)(51)
- Vaccinations against respiratory illness (eg, influenza, pneumococcal, respiratory syncytial virus, COVID-19)(1)
- Palliative care(1)(52)
- Discharge Planning as appropriate(2)(53)(54)
- Patient and caregiver education as appropriate. See Heart Failure: Patient Education for Clinicians [SR](#).(53)(54)

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	- Inpatient care in the home - Social Determinants of Health Assessment - Discharge planning	Clinical Indications met[G] - Tachypnea, edema, and dyspnea - Possible hypoxia	Initial bed rest	- Parenteral medications - Oral hydration - Oral diet	- ECG, CXR - Cardiac biomarkers, BNP - Oxygen - Digoxin level if previously prescribed - Electrolytes - Weigh	- IV diuretics[A] - Renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred)[D] - Beta-blocker[H] - Aldosterone antagonist - Sodium-glucose cotransporter-2 inhibitor[E]
2	- Inpatient care in the home - Social Determinants of Health Assessment	Hemodynamic stability Mental status at baseline No evidence of myocardial ischemia Cardiac rate and rhythm acceptable Oxygenation at baseline or improved Pulmonary edema absent or improved	Advance activity as tolerated	- Taper parenteral medications - Oral diet - Adjust fluid restriction	- Weigh - Possible electrolytes[I]	- Diuretics - Renin-angiotensin inhibitor - Beta-blocker - Aldosterone antagonist - Sodium-glucose cotransporter-2 inhibitor - Possible nitrates, hydralazine

3	• Social Determinants of Health Assessment • Inpatient care in the home to discharge • Complete discharge planning	• Hemodynamic stability • Tachypnea absent • Oxygenation at baseline or acceptable for next level of care • Dyspnea at baseline or acceptable for next level of care • Cardiac rate and rhythm acceptable • Pulmonary edema absent or acceptable for next level of care • Peripheral or sacral edema absent or acceptable for next level of care • Mental status at baseline • Volume status acceptable on oral medication • Renal function at baseline or acceptable for next level of care • Electrolyte abnormalities absent or acceptable for next level of care • Precipitating factors absent or controlled • Discharge plans and education understood	• Ambulatory or acceptable for next level of care	• Oral hydration[J] • Oral medications or regimen acceptable for next level of care • Oral diet or acceptable for next level of care	• Weigh	• Diuretics • Renin-angiotensin inhibitor • Beta-blocker • Aldosterone antagonist • Sodium-glucose cotransporter-2 inhibitor • Possible nitrates, hydralazine
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(1)(2)(24)(25)(56)(57)**N**

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 2 days

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Return to Hospital

- Return to the hospital may be necessary due to(1)(2)(34)(57):
 - Acute cardiac ischemia, confirmed or suspected as precipitant or consequence of heart failure
 - Hypotension that is severe (eg, mean arterial pressure[K] less than 65 mm Hg, lactic acidosis) or persistent despite treatment in the home(60)(61)(62)
 - See Hemodynamic Instability: Common Complications and Conditions [ISC guideline](#) as appropriate.
 - Worsening heart failure such that required treatment is beyond capabilities of Hospital-at-Home (eg, intra-aortic balloon pump, ventricular assist device, continuous intravenous diuretic, IV vasodilator, IV inotrope)

- Need for invasive monitoring (eg, pulmonary-artery catheter, arterial line)
- Need for telemetry monitoring with immediate onsite response capabilities (eg, dangerous arrhythmia or acute myocardial ischemia suspected)
 - See Arrhythmia: Common Complications and Conditions [ISC](#) guideline as appropriate.
- Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)
 - See Renal Failure: Common Complications and Conditions [ISC](#) guideline as appropriate.
- Need for initiation of invasive or noninvasive ventilatory assistance (eg, worsening hypoxemia)
 - See Respiratory Insufficiency: Common Complications and Conditions [ISC](#) guideline as appropriate.
- Altered mental status that is severe or persistent
 - See Mental Status Change: Common Complications and Conditions [ISC](#) guideline as appropriate.
- Development or worsening of secondary condition such that necessary care needs are beyond capabilities of Hospital-at-Home (eg, pneumonia, COPD, UTI, pulmonary embolus, psychiatric illness)
 - See Respiratory Insufficiency: Common Complications and Conditions [ISC](#), Urinary Complications: Common Complications and Conditions [ISC](#), Venous Thrombosis and Pulmonary Embolism: Common Complications and Conditions [ISC](#), Psychiatric Disorders: Common Complications and Conditions [ISC](#) guidelines as appropriate.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(1)(2)(34)(57):
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Stage IV chronic kidney disease (estimated glomerular filtration rate of less than 30 mL/min/1.73m² (0.50 mL/sec/1.73m²), and not previously on chronic dialysis) [eGFR - Adult Calculator\(63\)\(64\)](#)
 - Expect brief stay extension.
 - Concomitant pneumonia
 - Expect brief stay extension.
 - Significant electrolyte abnormality (hyponatremia, hypokalemia)
 - Expect brief stay extension.

See Common Complications and Conditions [ISC](#) for further information.

Hospital-at-Home Care Planning

- Hospital evaluation and care needs may include(1)(2)(26)(27)(34)(37)(62):[NN](#)
 - Diagnostic test scheduling and completion, including(38)(65):
 - CBC, electrolytes, urinalysis, digoxin level, kidney function, liver function, lipid panel, thyroid function, fasting glucose, coagulation profile, and glycosylated hemoglobin
 - Cardiac biomarkers (eg, BNP, N-terminal pro-BNP level, Troponin)(39)(40)(65)
 - Vitamin D level, iron studies(41)(66)
 - Treatment and procedure scheduling and completion, including:
 - Medication (diuretics, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), beta-blocker,^[H] aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide))^{[A][D][E](1)(27)}
 - Antihypertensive medication^{[F](29)}
 - Intravenous iron(27)(67)
 - Thromboembolic prophylaxis
 - Timely discontinuation of therapies (eg, switch to oral diuretics)
 - Consultation, assessment, and other services scheduling and completion, including:
 - Cardiology consultation
 - Cardiac rehabilitation consultation(50)(51)
 - Smoking cessation counseling(2)
 - Nutrition counseling(1)(29)
 - Sleep medicine consultation(29)(68)
 - Vaccinations against respiratory illness (eg, influenza, pneumococcal, respiratory syncytial virus, COVID-19)(1)
 - Palliative care consultation(1)(29)(52)
 - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool [SR](#)); key items include:
 - Complicating illness (eg, COPD, diabetes)(69)
 - Cardiovascular status, including:

- Cardiac rate and rhythm, blood pressure (including postural changes), and peripheral pulses
- Activity tolerance
- Oxygen requirements to maintain saturation of arterial oxygen greater than 90%
- Fluid balance
- Electrolytes and renal function
- Severity of peripheral or presacral edema
- Daily weight
- Presence of abdominal pain
- Depression(70)

Discharge

Discharge Planning

- Discharge planning includes[L]:
 - Assessment of needs and planning for care, including(72)(73):
 - Develop and modify treatment plan (involving multiple providers) as needed.
 - Evaluate and address preadmission functioning as needed.
 - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment [SR](#) for further information.
 - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool [SR](#) for further information.(71)
 - Evaluate and address patient or caregiver preferences as indicated.
 - Identify skilled services needed at next level of care, with specific attention to:
 - Cardiovascular status assessment(74)
 - Medication management, adherence instruction, and side effects assessment(75)
 - Nutrition and hydration management(76)
 - Ongoing education required(76)
 - Psychosocial assessment, management, and referrals(77)
 - Early identification of anticipated discharge destination; options include(73)(78):
 - Home, considerations include:
 - Home safety assessment. See Home Safety Assessment [SR](#) for further information.
 - Patient safe to go home
 - Access to follow-up care
 - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#) for further information.
 - Caregiver need, ability, and availability
 - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool [SR](#) for further information.(33)
 - Transition of care plan complete, including(73)(78)(79):
 - Patient and caregiver education complete
 - See Teach Back Tool [SR](#) for further information.
- ☐ Medication reconciliation completion includes(75)(80):
 - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
 - Assess each medication for correlation to disease state or medical condition.
 - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
 - Provide reconciled medication list to all treating providers.
 - Confirm that patient or caregiver can acquire medication.
 - Educate patient and caregiver.
 - Provide complete medication list to patient and caregiver.
 - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
 - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(81)
 - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
 - See Medication Reconciliation Tool [SR](#) for further information.
 - Plan communicated to patient, caregiver, and all members of care team, including(82)(83):
 - Inpatient care and service providers
 - Primary care provider
 - All postdischarge care and service providers

- Appointments planned or scheduled, which may include:
 - Primary care provider(84)
 - Cardiologist(85)
 - Dietitian(84)
 - Follow-up evaluation call. See Post-Acute Care Follow-Up Call Tool [SR](#) for more information.(86)
 - Palliative or end-of-life care provider(87)
 - Rehabilitation therapy services(84)
 - Specialists for management of comorbidities as needed(88)
 - Other: _____
- Outpatient testing and procedure plans made, which may include:
 - Echocardiogram
 - Exercise stress test(84)
 - Laboratory testing(89)
 - Other: _____
- Referrals made for assistance and support, which may include:
 - Alcohol and other drug abuse or dependence treatment(90)
 - Community services
 - Educational program (eg, chronic condition management, self-management)(89)(91)
 - End-stage disease planning(84)
 - Financial, for follow-up care, medication, and transportation
 - Protective services
 - Self-help or support groups(84)
 - Social services (eg, social programs, advance directives)(91)
 - Tobacco use treatment(84)(89)
 - Other: _____
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include(84):
 - Blood pressure cuff
 - Weight scale
 - Other: _____

Discharge Destination

- Post Hospital-at-Home (HaH) levels of admission may include:
 - Home.
 - Home healthcare. See Heart Failure [HC](#) guideline in Home Care.
 - Recovery facility care. See Heart Failure [RFC](#) guideline in Recovery Facility Care.

Evidence Summary

Background

The purpose of the Hospital-at-Home (HaH) guidelines is to assist in identification of patients in the emergency department who require inpatient care (ie, are "sick enough") but are appropriate for this inpatient care to be provided in the home (ie, not "too sick"). These guidelines presume that HaH care is for patients who truly need inpatient care (ie, absent an HaH program, the patient would be appropriate for inpatient admission in the hospital). Consequently, the same decision-making process should take place as is done for "brick and mortar" hospital inpatient admissions. For example, use of observation care may be appropriate before a decision to admit to inpatient care is necessary. The diagnosis-specific Observation Care guideline can be used in junction with either the Optimal Recovery Guideline or HaH guideline.

The Clinical Indications in the HaH guidelines focus upon the diagnosis-specific clinical presentation rather than an assessment of whether a given HaH program has the capabilities to provide the necessary care. These capabilities will vary across programs (eg, for imaging and laboratory testing). Each program will also have specific requirements in terms of the home environment, caregiver capabilities, and overall social support structure. These elements are not included in the HaH guidelines as it would not be possible to be specific enough to fit all programs. As is the case for all of the inpatient guidelines, there is the presumption that if appropriate capabilities are not present, HaH will not be an option. This presumption is not unique to HaH (eg, admission criteria for atrial fibrillation do not specify that appropriate telemetry services and monitoring are available). In the case of HaH, if a specific program is not able to provide the clinically required services (eg, frequency of medication dosing or nursing evaluations), then admission to the hospital rather than HaH is appropriate.

Criteria for HaH under the Centers for Medicare and Medicaid Services Acute Hospital Care at Home (AHCaH) waiver initiative include provision of selected essential hospital services (eg, pharmacy, infusion, oxygen delivery, nutritional support, social work, respiratory, physical, speech, and occupational therapy services), daily visits by a physician or advanced practitioner, capability for emergency personnel response within 30 minutes, at least twice-daily nursing or paramedical visits with vital sign monitoring, immediate remote

audio response capability, and safety reporting criteria.(10)(11) Patients eligible for HaH under the AHCaH waiver include only those seen in the emergency department (admission diversion) and those transferred from an inpatient hospital (early inpatient discharge). (10)(11) Select citations have been identified that may be useful in helping define the purpose, design, and utility of HaH.(5)(6)(12)(13)(14)(15)(16)(17) Also included are general citations (mixed diagnoses) that have detail or criteria concerning patient selection for HaH(5)(6)(13)(14)(15)(16) and citations specific to HaH for heart failure patients.(7)(8)(9) Updated information on program planning, financial and policy considerations, and examples of program eligibility criteria from individual hospital systems can be found on the HaH users group website at <https://www.hahusersgroup.org/>.(6)

Criteria

A meta-analysis of 16 randomized controlled trials (1814 patients) comparing Hospital-at-Home (HaH) care to acute "brick and mortar" hospitalization for a broad variety of conditions (including, but not limited to, COPD, stroke, cellulitis, community-acquired pneumonia, and heart failure) found no difference in 6-month mortality or need for hospital readmission; however, HaH patients had a lower likelihood of residing in a skilled nursing facility at 6-month follow-up.(18) **(EG 1)** A study of 1055 elderly United Kingdom patients (mean age 83 years) randomized to either HaH or to hospital-based inpatient care for a hemodynamically stable acute medical illness (including 45% infectious, 13% cardiovascular, and 8.5% respiratory diagnoses) found no differences in the outcome of living at home at 6 months (relative risk (RR) 1.05 (95% confidence interval (CI) 0.95 to 1.15)) and a reduced risk of long-term care placement at 6 months in the HaH group (RR 0.58 (CI 0.45 to 0.76)).(19) **(EG 1)** In the same study, readmission or transfer to a hospital at 1 month was higher in the HaH group (RR 1.32 (CI 1.06 to 1.64)), and while there was no difference detected in 6-month mortality, the confidence interval was wide (RR 0.98 (CI 0.65 to 1.47)); therefore, a significant increase or decrease in mortality cannot be excluded. (19) **(EG 1)** Analysis of 1878 patients admitted to HaH under the AHCaH waiver program found 7% required escalation to "brick and mortar" hospital care and reported an unexpected mortality rate of 0.4%.(11) **(EG 2)** A study of 172 patients (mean age 69 years) admitted to hospital at home (daily in-home nurse or paramedic visit, IV medications, home testing, and remote monitoring) for acute medical illness (infection, heart failure, or airway disease) and randomized to remote or in-home physician visits found that the remote visit group had noninferior outcomes for adverse events (6.8% vs 3.9% for remote and home visit, respectively; mean difference 2.8, 95% CI -3.3 to 8.9) and patient experience questionnaire scores.(20) **(EG 1)**

Alternatives

Analysis of 52,055,378 adult patients seen in the emergency department for heart failure over a 10-year period from the Nationwide Emergency Department Sample database reported 30% were treated without inpatient admission.(24) **(EG 2)** Analysis of 1,305,639 adult patients seen in the emergency department for heart failure from the Nationwide Inpatient Sample database reported 29% were treated without inpatient admission.(25) **(EG 2)** Analysis of 1756 pediatric patients (mean age 4.9 years) seen in the emergency department for heart failure from the Kids' Inpatient Database reported 25% were treated without inpatient admission.(25) **(EG 2)**

Hospitalization

A specialty guideline recommends 4 medication classes as guideline-directed medical therapy (GDMT) for patients with heart failure with reduced ejection fraction (HFrEF): beta-blockers, renin-angiotensin inhibitors (angiotensin receptor-neprilysin inhibitor preferred), aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors.(1) **(EG 2)** This same guideline recommends the following for patients hospitalized with HFrEF absent clear contraindications: pre-existing GDMT should be continued and optimized (titration to therapeutic dose begun as inpatient); GDMT should not be discontinued due to mild decrease in renal function or blood pressure; GDMT should be initiated while hospitalized when clinical stability achieved (if all drug classes not on board); and if GDMT is discontinued, it should be reinstated as soon as possible.(1) **(EG 2)** An expert panel also recommends these 4 medication classes as GDMT for heart failure with preserved ejection fraction (HFpEF).(29) **(EG 2)**

A specialty clinical guideline based upon a systematic review of the evidence concludes that heart failure patients with evidence of significant fluid overload should be promptly treated with intravenous loop diuretics (eg, initial dose twice the home regimen in terms of oral to parenteral milligram equivalents) and titrated to resolution of clinical signs and symptoms of congestion (eg, serial doubling of dosage as bolus or continuous infusion).(1) **(EG 2)** This guideline further recommends increasing the diuretic dose or adding a second diuretic (eg, thiazide diuretic or aldosterone antagonist with diuretic properties) until an effective regimen is identified that is adequate to relieve congestion.(1) **(EG 2)**

Length of Stay

Analysis of 31,137,414 patients admitted with heart failure from the Nationwide Inpatient Sample database over a 10-year period found that 25% of patients were discharged in 2 days or less.(24) **(EG 2)** Analysis of 950,045 adult patients admitted with heart failure from the Nationwide Inpatient Sample database found that 25% of patients were discharged in 2.1 days or less.(25) **(EG 2)** A study including 978,135 patients (mean age 72 years) hospitalized with a primary diagnosis of heart failure found that 25% of patients were discharged within 2 days.(56) **(EG 2)** Analysis of national hospital discharge data shows 23% of hospitalized patients with the primary diagnosis of heart failure discharged in 2 days or less.(57) **(EG 3)**

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Footnotes

[A] A specialty clinical guideline based upon a systematic review of the evidence concludes that heart failure patients with evidence of significant fluid overload should be promptly treated with intravenous loop diuretics (eg, initial dose twice the home regimen in terms of oral to parenteral milligram equivalents) and titrated to resolution of clinical signs and symptoms of congestion (eg, serial doubling of dosage as bolus or continuous infusion).(1) This guideline further recommends increasing the diuretic dose or adding a second diuretic (eg, thiazide diuretic or aldosterone antagonist with diuretic properties) until an effective regimen is identified that is adequate to relieve congestion.(1) [A in Context Link 1, 2, 3, 4]

[B] The Get With The Guideline Heart Failure (GWTG-HF) Risk Score is a clinical prediction tool that has been validated to estimate risk for in-hospital mortality in hospitalized heart failure patients.(21)(22) A risk estimate of less than 10% has been proposed for use in selecting heart failure patients who may be appropriate for Hospital-at-Home.(13) An online calculator is available at <https://www.mdcalc.com/calc/3829/gwtg-heart-failure-risk-score>. [B in Context Link 1, 2]

[C] While some telemetry monitoring may be performed in Hospital-at-Home, if a dangerous arrhythmia is detected (eg, ventricular arrhythmia), response would need to be immediate.(10) [C in Context Link 1]

[D] A specialty guideline recommends 4 medication classes as guideline-directed medical therapy (GDMT) for patients with heart failure with reduced ejection fraction (HFrEF): beta-blockers, renin-angiotensin inhibitors (angiotensin receptor-neprilysin inhibitor preferred), aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors.(1) This same guideline recommends the following for patients hospitalized with HFrEF absent clear contraindications: pre-existing GDMT should be continued and optimized (titration to therapeutic dose begun as inpatient), GDMT should not be discontinued due to mild decrease in renal function or blood pressure, GDMT should be initiated while hospitalized when clinical stability achieved (if all drug classes not on board), and if GDMT is discontinued it should be reinstated as soon as possible.(1) Vericiguat further reduces the risk of adverse cardiac outcomes in heart failure and may either be started with the standard quadruple therapy outlined above for severe worsening heart failure, used as a substitute agent for patients with contraindication or intolerance to one or more standard medications, or used as an add-on therapy for inadequate response to optimal quadruple therapy.(27) An expert specialty society panel also recommends these 4 medication classes as GDMT for heart failure with preserved ejection fraction (HFpEF).(29) [D in Context Link 1, 2, 3, 4]

[E] Specialty guidelines recommend sodium-glucose cotransporter-2 inhibitors for nondiabetic and type 2 diabetic patients with heart failure because they reduce the risk of worsening heart failure, major adverse cardiac events, and cardiac mortality.(1)(44) However, they should be avoided in severe illness, during prolonged fasting, or for surgical procedures.(44) They are not currently FDA approved for patients with type 1 diabetes due to risk of ketoacidosis.(1) [E in Context Link 1, 2, 3]

[F] Additional hypertensive agents beyond the recommended heart failure guideline-directed medical therapy (GDMT) medications may be needed for patients with heart failure with reduced ejection fraction (HFpEF) to attain a goal systolic blood pressure less than 130 mm Hg.(29) [F in Context Link 1, 2]

[G] See Clinical Indications for Admission to Hospital-at-Home in this guideline. [G in Context Link 1]

[H] Beta-blockers may not be appropriate in the setting of acute pulmonary edema.(1) Recommended beta-blockers include carvedilol, bisoprolol, and extended-release metoprolol.(1)(55) [H in Context Link 1, 2]

[I] Electrolyte monitoring may be indicated if patient is actively diuresing.(1) [I in Context Link 1]

[J] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [J in Context Link 1]

[K] The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals $1/3 \text{ SBP} + 2/3 \text{ DBP}$.(58)(59) [K in Context Link 1]

[L] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(71) [L in Context Link 1]

Definitions

Acute kidney injury (stage 2)

- Acute kidney injury (stage 2) with significant clinical findings, as indicated by **ALL** of the following(1)(2)(3):
 - Acute[A] kidney injury (stage 2), as indicated by **1 or more** of the following[B]:
 - Within past 7 days, rise in serum creatinine between 2 and 2.9 times its baseline value (increase of 3 times baseline would qualify as stage 3 acute kidney injury)
 - Decreased urine output,[C] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Urine output less than 0.5 mL/kg/hr for 12 hours
 - Significant clinical findings, as indicated by **1 or more** of the following[B]:
 - Hemodynamic instability
 - Worsening clinical status despite observation care (eg, rising creatinine or urine output remains less than 0.5 mL/kg/hr)
 - Altered mental status
 - Cardiac arrhythmias of immediate concern
 - Severe hypertension (SBP greater than 180 mm Hg or DBP greater than 120 mm Hg in adults or greater than the 95th percentile for age, gender, and height plus 30 mm Hg in pediatric patients)(4)(5)
 - Significant respiratory findings (eg, pulmonary edema) that are severe (eg, Hypoxemia) or persist despite observation care
 - Clinically significant electrolyte abnormality that is severe (eg, hyperkalemia with severe ECG findings)[D] or persists despite observation care
 - Clinically significant metabolic abnormality (eg, metabolic acidosis) that is severe or persists despite observation care

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Footnotes

- A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- B. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- C. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)
- D. ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(6)(7)

Acute renal failure (stage 3 acute kidney injury)

- Acute^[A] kidney injury (stage 3),^[B] as indicated by **1 or more** of the following^[C](1)(2)(3):
 - Within past 7 days, rise in serum creatinine to 3 times its baseline value or higher
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%) and serum creatinine is greater than 4 mg/dL (354 micromoles/L)
 - In child younger than 18 years, estimated glomerular filtration rate less than 35 mL/min/1.73m² (0.59 mL/sec/1.73m²)
-  eGFR - Pediatric Calculator and **1 or more** of the following:
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%)
 - Within past 48 hours, rise in serum creatinine greater than 0.3 mg/dL (26.5 micromoles/L)
- Decreased urine output,^[D] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Oligoanuria, as indicated by **1 or more** of the following:
 - Urine output less than 0.3 mL/kg/hour for 24 hours
 - Anuria (urine output less than 0.1 mL/kg/hour) for 12 hours
- Initiation of renal replacement therapy (RRT) required(4)

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Footnotes

- A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- B. For purposes of these criteria, the term acute renal failure can be viewed as stage 3 acute kidney injury in the KDIGO classification system.(1) Of note, in the ICD-10 diagnostic coding system (main coding scheme used in labeling diagnoses), the relevant codes covering the clinical entity of significant acute kidney injury use the term acute kidney failure (eg, code N17.9).
- C. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- D. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear

relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:117-125.e1.
3. Abraham G, Maher PJ. Delirium and dementia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1269-1280.e2.
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Altered mental status that is severe or persistent

- Altered mental status (ie, different from baseline) that is severe or persistent, as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:117-125.e1.
3. Abraham G, Maher PJ. Delirium and dementia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1269-1280.e2.
4. Schor NF. Neurologic evaluation. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:3053-3063.e1.

Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)
 - Nonsustained ventricular tachycardia and **1 or more** of the following:
 - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
 - Acute myocarditis
 - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Any heart rhythm and **1 or more** of the following(4)(5)(9)(10)(11):
 - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)
 - Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.

- Heart rhythms of concern due to **1 or more** of the following:
 - Hypotension
 - Respiratory distress
 - Association with other significant symptoms (eg, syncope)(9)(10)(12)

References

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Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
 - Vital sign abnormality not readily corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
 - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
 - Hypotension that persists despite appropriate treatment (eg, volume repletion, treatment of underlying cause)
 - Orthostatic hypotension that persists despite appropriate treatment (eg, volume repletion)
 - Hypotension that is severe, as indicated by **ALL** of the following:
 - Hypotension
 - Severe hypotension, as indicated by **1 or more** of the following:
 - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
 - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
 - Mean arterial pressure[A][D] less than 65 mm Hg
 - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)
- C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated. (12)
- D. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals 1/3 SBP + 2/3 DBP.(1)(4) A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

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Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older^[A]
 - Mean arterial pressure^{[A][B]} less than 70 mm Hg in adult or child 10 years or older
 - Decrease in baseline mean arterial pressure of 25% or more,^{[A][B]} with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
 - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age^[A]
 - SBP less than 70 mm Hg in infant 1 to 11 months of age^[A]

References

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent^[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
 - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)[A]
 - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

References

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- Rojas-Camayo J, et al. Reference values for oxygen saturation from sea level to the highest human habitation in the Andes in acclimatised persons. Thorax 2018;73(8):776-778. DOI: 10.1136/thoraxjnl-2017-210598.

Footnotes

- A. Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

- Shibao C, Lipsitz LA, Biaggioni I, American Society of Hypertension Writing Group. Evaluation and treatment of orthostatic hypotension. Journal of the American Society of Hypertension 2013 Jul-Aug;7(4):317-324. DOI: 10.1016/j.jash.2013.04.006.
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Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)

- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Patient safe to go home

- Patient safe to go home; examples include(1)(2)(3):
 - Medical status stable for patient's condition
 - Functional care can safely be provided with available resources.
 - Mental status stable for patient's condition
 - Medication availability confirmed and reconciliation complete
 - Patient/caregiver education completed with written discharge instructions provided
 - Community resources identified and referrals made, as needed
 - Home care arranged, if indicated
 - Necessary medical equipment delivery arranged or available in home, if indicated
 - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated

References

1. Elements of Excellence in Transitions of Care (TOC). TOC Checklist [Internet] National Transitions of Care Coalition. Accessed at: <https://www.ntocc.org>. [accessed 2023 Aug 10]
2. Medical-surgical nursing. In: Hinkle JL, Cheever KH, Overbaugh KJ, editors. Brunner & Suddarth's Textbook of Medical-Surgical Nursing. 15th ed. Wolters Kluwer; 2022:33-55.
3. Transitions of care. In: Gillingham DC, editor. Foundations of Case Management. Blue Bayou Press; 2021:137-144.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
 - Patient with **1 or more** of the following:
 - Dyspnea (difficulty breathing)
 - Tachypnea
 - Abnormal breathing pattern (eg, chest retractions)
 - Other evidence of difficulty breathing
 - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
 - Hypoxemia
 - Altered mental status
 - Other evidence of respiratory compromise (eg, respiratory acidosis)

References

1. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
2. Naureckas ET, Solway J. Disturbances of respiratory function. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2133-2139.

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
 - Housing insecurity, as indicated by **1 or more** of the following:
 - Individual or caregiver's current living situation is **1 or more** of the following(5):
 - Does not have own housing (eg, staying in a hotel, shelter, or with others)
 - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
 - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
 - Current housing has **1 or more** of the following:
 - Electrical appliances (eg, stove, refrigerator) not working or unavailable
 - Insufficient heating or cooling
 - Insufficient ventilation
 - Lead paint or pipes
 - Mold
 - Pests (eg, bugs) or rodents

- Smoke detectors not working or unavailable
- Food insecurity, as indicated by **1 or more** of the following(6):
 - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
 - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
- Insufficient transportation, as indicated by **1 or more** of the following(7):
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
 - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
- Insufficient utilities, as indicated by **1 or more** of the following(8):
 - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
 - In the past year, electric, water, gas, or oil company threatened to shut off services.
- Personal safety risk, as indicated by **2 or more** of the following(6):
 - Individual is sometimes or frequently physically hurt by another person (including family member).
 - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
 - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
 - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
- Insufficient dependent care, as indicated by **1 or more** of the following:
 - In the past year, individual or caregiver was unable to work due to lack of dependent care.
 - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
 - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
 - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
 - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

References

1. Scanlon A, Reinisch C. Social determinants of health. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. *Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems*. 12th ed. St. Louis, MO: Mosby; 2023:18-20.
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Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
 - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
 - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
 - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
 - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
 - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
 - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.

2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia^[A]_[B] absent, as indicated by **1 or more** of the following⁽¹⁾₍₂₎:
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
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Tachypnea

- Tachypnea^[A]_[B] as indicated by respiratory rate of **1 or more** of the following⁽²⁾₍₃₎:
 - Greater than 18 breaths per minute in adult or child 13 years of age or older^[A]
 - Greater than 22 breaths per minute in child 6 to 12 years of age^[A]
 - Greater than 25 breaths per minute in child 3 to 5 years of age^[A]
 - Greater than 30 breaths per minute in child 1 or 2 years of age^[A]
 - Greater than 40 breaths per minute in infant 6 to 11 months of age^[A]
 - Greater than 45 breaths per minute in infant 3 to 5 months of age^[A]
 - Greater than 60 breaths per minute in infant 1 or 2 months of age^[A]

References

1. Schrager DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
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- B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(1) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(1)

Tachypnea absent

- Tachypnea[A][B] absent, as indicated by respiratory rate of **1 or more** of the following(2)(3):
 - Less than or equal to 18 breaths per minute in adult or child 13 years of age or older
 - Less than or equal to 22 breaths per minute in child 6 to 12 years of age
 - Less than or equal to 25 breaths per minute in child 3 to 5 years of age
 - Less than or equal to 30 breaths per minute in child 1 or 2 years of age
 - Less than or equal to 40 breaths per minute in infant 6 to 11 months of age
 - Less than or equal to 45 breaths per minute in infant 3 to 5 months of age
 - Less than or equal to 60 breaths per minute in infant 1 or 2 months of age

References

1. Schrager DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
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Troponin

- Troponin elevation alone does not indicate acute myocardial infarction; factors to consider include(1)(2):
 - Patients can have elevated levels of cardiac troponin due to other causes of myocardial injury, either acutely or chronically (eg, heart failure, myocarditis, pulmonary embolism, cardiomyopathy, defibrillator shock).
 - Patients with an elevated troponin should, on subsequent measurement, have a fall or rise in the troponin level of more than 20% to confirm acuity of myocardial injury. Absent this variation, the acuity of the myocardial injury should be questioned.
 - Attributing acute troponin release to myocardial ischemia depends on the presence of associated symptoms (eg, chest pain, dyspnea), ECG changes (eg, acute ST-segment deviation), and imaging findings (eg, new wall motion abnormalities on echocardiogram).
 - A report from a facility (eg, hospital) laboratory of an elevated troponin level, in general, indicates a level above the 99th percentile for the specific troponin assay used. The precise value that represents the 99th percentile depends on the specific troponin assay used.
 - The recommended time interval between serial troponin level assessments (eg, to detect fall or rise in troponin level) is 1 to 3 hours for high-sensitivity troponin, and 3 to 6 hours for conventional (contemporary) troponin assays.
 - Evidence-based specialty society guidelines recommend use of high-sensitivity troponin assays, rather than older less-sensitive assays (collectively labeled contemporary assays).(1)(3) This is due to more rapid detection or exclusion of myocardial injury and increased diagnostic accuracy with high-sensitivity assays.(1)(3)

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Codes

ICD-10 Diagnosis: I09.81, I11.0, I13.0, I13.2, I50.1, I50.20, I50.21, I50.22, I50.23, I50.30, I50.31, I50.32, I50.33, I50.40, I50.41, I50.42, I50.43, I50.814, I50.82, I50.83, I50.84, I50.89, I50.9, T86.22 [Hide]

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